# SATELLITE-BASED ATTRIBUTION OF ANTHROPOGENIC STORAGE SIGNALS IN DAMODAR VALLEY CORPORATION RESERVOIRS USING EXPLAINABLE MACHINE LEARNING AND CAUSAL INFERENCE

*A Project Report*

*Submitted to*

**JHARKHAND UNIVERSITY OF TECHNOLOGY, RANCHI**

*In Partial Fulfillment of the Requirement for the Award of Degree of*

**BACHELOR OF TECHNOLOGY IN CIVIL ENGINEERING**


Submitted by

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Under the guidance of

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**June 2026**

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**June 2026**

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## CERTIFICATE OF DECLARATION FROM THE SUPERVISOR

This is to certify that the work presented in the dissertation entitled **"Satellite-Based Attribution of Anthropogenic Storage Signals in Damodar Valley Corporation Reservoirs Using Explainable Machine Learning and Causal Inference"** in partial fulfillment of the requirement for the award of Degree of **Bachelor of Technology** in **Civil Engineering** at **BIT Sindri** is an authentic work carried out under my supervision and guidance.

To the best of my knowledge, the content of this thesis does not form a basis for the award of any previous Degree to anyone else.

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## CERTIFICATE OF APPROVAL FROM THE EXAMINERS

The foregoing dissertation entitled **"Satellite-Based Attribution of Anthropogenic Storage Signals in Damodar Valley Corporation Reservoirs Using Explainable Machine Learning and Causal Inference"**, is hereby approved as a creditable study of research topic and has been presented in satisfactory manner to warrant its acceptance as prerequisite to the degree for which it has been submitted.

It is understood that by this approval, the undersigned do not necessarily endorse any conclusion drawn or opinion expressed therein, but approve the thesis for the purpose for which it is submitted.

_____ _____

**(Internal Examiner)** **(External Examiner)**

_____

**(Chairman)**

**Head of the Department**

# DECLARATION

I hereby declare that the project work entitled **"Satellite-Based Attribution of Anthropogenic Storage Signals in Damodar Valley Corporation Reservoirs Using Explainable Machine Learning and Causal Inference"** is an authentic record of my own work carried out during the academic year 2025–2026 as a requirement for the award of the degree of Bachelor of Technology in Civil Engineering from BIT Sindri, Dhanbad.

The work presented in this thesis has not been submitted to any other university or institute for the award of any degree or diploma.

All sources of information have been duly acknowledged.

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**Nishi Kant Mandal**

Sindri, June 2026

# ABSTRACT

Reservoir operations are a dominant but poorly understood source of terrestrial water storage variability in managed river basins. Satellite missions like the Gravity Recovery and Climate Experiment Follow-On (GRACE-FO) measure total water storage changes at basin scale, yet they cannot distinguish between climate-driven processes and deliberate human management decisions at individual reservoirs. This matters because water policy in India increasingly relies on GRACE-derived estimates of groundwater depletion and water stress. In basins where reservoir operations contribute substantially to the observed storage signal, these estimates may be systematically biased.

We developed a six-stage satellite-based framework to isolate and characterise anthropogenic storage signals in the Damodar Valley Corporation (DVC) reservoir cascade in Jharkhand and West Bengal, using Sentinel-1 C-band Synthetic Aperture Radar (SAR) imagery, GRACE-FO mascon data, and climate reanalysis products. We reconstructed monthly storage anomalies for Maithon Dam and Durgapur Barrage over 2017–2024 and validated them against nine independent measurements from the Central Water Commission (CWC; RMSE = 0.07 m, $n = 8$). An eXtreme Gradient Boosting (XGBoost) model trained on eight climate drivers achieved modest predictive skill (Nash-Sutcliffe Efficiency, NSE = 0.208 on unclipped rows). Climate alone explains very little of month-to-month storage variability in a managed reservoir. The anthropogenic residual ranged from $-0.19$ to $+0.23$ Billion Cubic Metres (BCM). Convergent Cross-Mapping (CCM) with 999 phase-randomised surrogates confirmed that this residual is causally independent of all climate predictors. Partial regression against GRACE-FO showed that the anthropogenic signal remains detectable even after removing shared monsoon seasonality ($p = 0.0003$). A Dynamic Time Warping (DTW) based operational fingerprint taxonomy (silhouette = 0.49) showed that DVC operational patterns can be learned from satellite data alone, without any in-situ sensors.

**Keywords:** reservoir operations, Damodar Valley Corporation, Sentinel-1 SAR, GRACE-FO, XGBoost, SHapley Additive exPlanations, convergent cross-mapping, operational fingerprint taxonomy

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# List of Abbreviations

| | |
|---|---|
| ACF | Autocorrelation Function |
| ADF | Augmented Dickey-Fuller (stationarity test) |
| AET | Actual Evapotranspiration |
| ALOS | Advanced Land Observing Satellite |
| API6 | 6-month Antecedent Precipitation Index |
| AV | Area–Volume (curve) |
| AW3D30 | ALOS World 3D 30-metre DEM |
| BCM | Billion Cubic Metres |
| CCM | Convergent Cross-Mapping |
| CHIRPS | Climate Hazards Group InfraRed Precipitation with Station data |
| CN | Curve Number (SCS method) |
| CWC | Central Water Commission |
| DEM | Digital Elevation Model |
| DMI | Dipole Mode Index (Indian Ocean Dipole) |
| DTW | Dynamic Time Warping |
| DVC | Damodar Valley Corporation |
| DW | Durbin-Watson (autocorrelation statistic) |
| ECMWF | European Centre for Medium-Range Weather Forecasts |
| ENSO | El Niño–Southern Oscillation |
| ERA5 | ECMWF Reanalysis Fifth Generation |
| FSL | Full Supply Level |
| GEE | Google Earth Engine |
| GRACE-FO | Gravity Recovery and Climate Experiment Follow-On |
| GRD | Ground Range Detected (Sentinel-1 product type) |
| GWS | Groundwater Storage |
| HAC | Heteroscedasticity and Autocorrelation Consistent |
| HMM | Hidden Markov Model |
| IOD | Indian Ocean Dipole |
| JRC | Joint Research Centre (European Commission) |
| KGE | Kling-Gupta Efficiency |
| MODIS | Moderate Resolution Imaging Spectroradiometer |
| NDVI | Normalised Difference Vegetation Index |
| NDWI | Normalised Difference Water Index |
| NSE | Nash-Sutcliffe Efficiency |
| OLS | Ordinary Least Squares |

| | |
|---|---|
| PACF | Partial Autocorrelation Function |
| PBIAS | Percentage Bias |
| PET | Potential Evapotranspiration |
| SAR | Synthetic Aperture Radar |
| SCS-CN | Soil Conservation Service Curve Number |
| SHAP | SHapley Additive exPlanations |
| SMS | Soil Moisture Storage |
| STL | Seasonal-Trend decomposition using Loess |
| SWS | Surface Water Storage |
| TPE | Tree-structured Parzen Estimator (Optuna sampler) |
| TWS | Terrestrial Water Storage |
| VIF | Variance Inflation Factor |

# List of Symbols

| Symbol | Meaning | Units |
|---|---|---|
| $V$ | Reservoir storage volume | BCM |
| $A$ | Reservoir water surface area | $\mathrm{km}^2$ |
| $a, b$ | Area–volume power-law coefficients | — |
| $\Delta S_{\mathrm{obs}}$ | Observed storage anomaly | BCM |
| $\Delta S_{\mathrm{climate}}$ | Climate-predicted storage anomaly | BCM |
| $\Delta S_{\mathrm{anthro}}$ | Anthropogenic storage residual | BCM |
| $P(t)$ | Monthly precipitation | mm |
| $\mathrm{API}_6$ | 6-month antecedent precipitation index | mm |
| PET | Potential evapotranspiration | mm/month |
| AET | Actual evapotranspiration | mm/month |
| $\mathrm{SM}_1$ | Soil moisture, 0–7 cm layer | $\mathrm{m}^3/\mathrm{m}^3$ |
| $\mathrm{SM}_2$ | Soil moisture, 7–28 cm layer | $\mathrm{m}^3/\mathrm{m}^3$ |
| $\varrho$ | CCM cross-map correlation skill | — |
| $E$ | Embedding dimension (CCM/Simplex) | — |
| $L$ | Library size (CCM) | — |
| $\beta_0 \dots \beta_4$ | OLS regression coefficients | various |
| $R^2$ | Coefficient of determination | — |
| $S$ | SCS-CN maximum potential retention | mm |
| $I_a$ | SCS-CN initial abstraction | mm |
| $Q$ | Surface runoff depth | mm |
| $CN$ | SCS Curve Number | — |

# Chapter 1

# Introduction

## 1.1 Background and Motivation

India operates over 5,000 large dams and runs some of the most complicated multi-reservoir systems anywhere in the world. Yet our ability to actually track what these reservoirs do on a month-to-month basis is poor. Most reservoirs still report storage levels through manual gauge readings, compiled into weekly bulletins by the Central Water Commission (CWC). The process is slow, the spatial coverage is thin, and access is largely restricted to government agencies with the right institutional ties.

Satellite remote sensing has partially filled this gap. The Gravity Recovery and Climate Experiment (GRACE) and its successor GRACE Follow-On (GRACE-FO) have been measuring monthly Terrestrial Water Storage (TWS) anomalies at basin to continental scales since 2002 [1]. In India, GRACE data have been used to estimate groundwater depletion in the Indo-Gangetic Plain [2], assess water stress across states, and feed into policy frameworks for sustainable water use. These are useful applications. They do, however, depend on an assumption that rarely gets scrutinized: that the GRACE signal over a given region can be cleanly separated into groundwater, soil moisture, snow, and surface water, and that the surface water part responds mainly to climate.

In heavily managed basins, that assumption breaks down. A reservoir that releases 0.2 billion cubic metres (BCM) of storage in October, not because of drought but because the operating authority needs flood cushion ahead of the next monsoon, produces a TWS anomaly that looks identical, at GRACE resolution, to 0.2 BCM of groundwater loss from agricultural pumping. Without a way to tell the two apart, every GRACE-based estimate of groundwater depletion in a managed basin risks being contaminated by operational reservoir cycles that have nothing to do with aquifer health.

The scale of this matters. Indian reservoirs hold over 250 BCM of combined live storage. Pre-monsoon drawdowns and post-monsoon conservation filling generate TWS anomalies between $-0.1$ and $-0.5$ BCM per reservoir per month [3]. GRACE-based analyses may attribute these signals to climate-forced groundwater or soil moisture changes when they are actually dam operations. If even a small portion of the apparent depletion across Indian river basins turns out to be reservoir

management rather than aquifer decline, current water policy could be directing resources at the wrong problem.

This project started from a direct question: can we identify the fingerprint of reservoir operations using satellite data alone, and can we show that the signal is genuinely anthropogenic rather than unexplained climate variability?

## 1.2 The Damodar Valley: History and Significance

The Damodar River flows about 541 km from the Chota Nagpur Plateau down to where it meets the Hooghly near Kolkata. People have called it the "Sorrow of Bengal" for centuries, and for good reason. Around 80% of the catchment's 1,300 mm annual rainfall arrives during the June–September southwest monsoon. The upper catchment is steep, the soils are impermeable laterite, and the lower floodplain is flat. Put those together and you get catastrophic floods. The historical record has major ones in 1913, 1935, 1943, 1956, 1978, 2000, and 2009, all of which inundated large parts of the lower Damodar floodplain in West Bengal.

The Damodar Valley Corporation (DVC) was established by Act of Parliament in 1948, modelled on the Tennessee Valley Authority in the United States. This was barely a year after independence, and the DVC was among India's first multipurpose river valley projects. The idea was Nehruvian in spirit: large infrastructure as a vehicle for national development. The Corporation built a system of dams and barrages across the Damodar catchment. Three of these are central to the present study:

1. **Maithon Dam** (commissioned 1957): Located at 23.835°N, 86.821°E in Dhanbad district, Jharkhand. Design gross capacity 1.360 BCM, current estimated capacity 1.115 BCM after 18% sedimentation loss [4]. Functions as the primary flood control structure with hydropower generation (60 MW installed capacity). The reservoir has a full supply level (FSL) of approximately 155.5 m and a dead storage level below which water cannot be released by gravity.

2. **Panchet Dam** (commissioned 1959): Located at 23.577°N, 86.769°E, approximately 30 km southwest of Maithon. Design capacity 0.290 BCM. Operates in tandem with Maithon for flood control, with coordinated pre-monsoon drawdowns and emergency release protocols managed by the DVC's central control room.

3. **Durgapur Barrage** (commissioned 1955): Located at 23.487°N, 87.307°E in Paschim Bardhaman district, West Bengal, approximately 80 km downstream

of Maithon. This is a run-of-river structure (not a storage dam) that diverts water into the 2,494 km Durgapur Main Canal system, providing gravity-fed irrigation to approximately 400,000 hectares of Rabi cropland in West Bengal. The barrage's operational schedule is tied to the Rabi wheat sowing calendar (October–March), producing a rigid and predictable seasonal drawdown pattern.

The 2009 flood is the most important event in recent DVC history. Between 22 and 25 September, extreme rainfall across the upper catchment forced simultaneous emergency releases from both Maithon and Panchet, flooding downstream areas and displacing hundreds of thousands of people [5]. What the event showed is that the DVC system can reduce flood peaks but cannot eliminate them entirely. More relevant for this work, operational decisions during such events produce a large storage signal that has nothing to do with climate variability.

The DVC system works well as a study area because it is well documented and physically diverse. Maithon is a large storage reservoir with a fill-drawdown cycle that spans several months. Durgapur is a run-of-river barrage that follows a fixed irrigation schedule. These two structures operate very differently. If our methods can tell them apart using satellite data alone, that already demonstrates that satellite-based operational monitoring is feasible for India's many ungauged reservoirs.

## 1.3 Problem Statement

Three gaps in the existing literature motivate this work.

1. **The attribution gap.** Studies based on the Gravity Recovery and Climate Experiment (GRACE) routinely attribute Terrestrial Water Storage (TWS) changes to groundwater depletion or climate-forced soil moisture dynamics, but they rarely consider that managed reservoir operations may contribute substantially to the observed signal [2, 6]. To our knowledge, no study has isolated a reservoir-specific anthropogenic component at the scale of an individual Indian reservoir system with multi-method causal validation.

2. **The monitoring gap.** Reservoir monitoring in India still depends on manual gauge readings and institutional data-sharing arrangements that frequently break down. No operational framework exists for classifying reservoir regimes from passive satellite observations alone, even though such a capability would benefit the hundreds of ungauged or data-sparse reservoirs across the subcontinent.

3. **The causal gap.** Satellite-based attribution studies typically rely on statistical decomposition (regression or machine learning residuals) and do not apply

formal causal inference tests to check whether the estimated anthropogenic residual is genuinely independent of climate forcing [7]. A residual from a poorly specified model is not an anthropogenic signal; it is noise. Without causal validation the attribution is not credible.

## 1.4 Research Gap

Satellite reservoir monitoring, GRACE TWS analysis, explainable machine learning, and causal inference have each reached a reasonable level of maturity on their own, yet no single study has brought them together for an individual Indian reservoir system. Specifically, no existing work has:

(a) isolated the anthropogenic storage residual at reservoir management resolution using explainable machine learning with formal uncertainty quantification;

(b) verified its causal independence from climate using both linear (Granger) and nonlinear (Convergent Cross-Mapping) tests with phase-randomised surrogate validation;

(c) shown that this residual remains detectable in GRACE TWS after removing the shared monsoon seasonality; and

(d) built a satellite-only operational fingerprint taxonomy that classifies reservoir regimes without any in-situ data.

We address this combined gap using the Damodar Valley Corporation (DVC) system as a case study. Because the entire framework runs on open-source tools and freely available data, it can be reproduced at zero financial cost and transferred directly to other Indian institutions studying managed basins.

## 1.5 Research Objectives

The specific objectives of this study are:

(a) To isolate the anthropogenic storage residual in DVC reservoirs by training a climate-only machine learning model and verifying that the residual is statistically independent of all retained climate predictors.

(b) To develop a satellite-based operational fingerprint taxonomy using Dynamic Time Warping (DTW) clustering that can classify reservoir operational regimes from Synthetic Aperture Radar (SAR) data alone, without in-situ sensors or operator-reported discharge records.

(c) To assess whether the reservoir-scale anthropogenic storage anomaly explains a significant fraction of GRACE Follow-On (GRACE-FO) TWS variability in the Damodar mascon after controlling for climate forcing.

Each objective has a clearly defined pass criterion and a specific statistical test, described in Chapter 4. The objectives are ordered by increasing ambition: the first requires only that we can separate the signal, the second requires that we can classify it, and the third requires that we can detect it at a spatial scale 4× larger than the basin itself.

## 1.6 Scope and Limitations

We restrict our analysis to the Sentinel-1 era (2017–2024, 82 months) for Maithon Dam and Durgapur Barrage. Panchet Dam was excluded because the original Google Earth Engine (GEE) extraction bounding box captured the Damodar River floodplain and DVC canal network, contaminating 57% of records with physically impossible water areas (140–370 $km^2$ versus the expected 40–70 $km^2$). The MODIS era (2002–2013) was available but could not be cross-calibrated to SAR due to a four-year sensor gap (2014–2016 when Sentinel-1 had sparse revisit coverage over India).

All data sources used in this project are freely available. No proprietary software, commercial satellite imagery, or paid subscriptions were used at any stage. The total monetary cost of the project was zero rupees — a deliberate constraint that demonstrates the accessibility of the framework for resource-limited institutions like our own.

## 1.7 Organisation of the Thesis

This thesis is organised into six chapters.

Chapter 2 reviews the relevant literature in four areas: satellite-based reservoir monitoring, GRACE TWS analysis, machine learning in hydrology, and causal inference in Earth sciences.

Chapter 3 describes the Damodar River basin, the DVC reservoir system, all satellite and reanalysis datasets we used, and the validation data sources.

Chapter 4 presents the six-stage methodology in full detail. We include all equations, parameter choices, and algorithmic steps so that a reader with access to Google Earth Engine and Python can reproduce the analysis.

Chapter 5 reports the results of each stage, interprets them physically, places them in context with prior work, and gives an honest assessment of limitations

and uncertainties.

Chapter 6 summarises the principal conclusions, states each objective's outcome, and suggests specific directions for future work.

# Chapter 2

# Literature Review

## 2.1 Satellite Remote Sensing for Reservoir Monitoring

### 2.1.1 Optical Sensors and Their Limitations

Satellite monitoring of inland water bodies has expanded considerably since the early 2000s. Landsat (30 m, 16-day revisit) and the Moderate Resolution Imaging Spectroradiometer (MODIS, 250–500 m, daily) were the first sensors to produce global surface water maps. Pekel et al. [8] processed the full 32-year Landsat archive to create the Joint Research Centre (JRC) Global Surface Water dataset, which maps water occurrence, seasonality, and change at 30 m for the entire globe. Most reservoir studies now use this dataset as their reference water mask.

Optical sensors, though, depend on visible and near-infrared wavelengths that clouds block entirely. For the Damodar basin this is a serious problem. The June–September monsoon is exactly when reservoirs go through their most important operational changes (filling, emergency flood releases, peak storage), and cloud cover in those months routinely exceeds 80%. We confirmed this directly: MODIS MOD09GA returned 0.000 $km^2$ water area for every single June–September month from 2002 to 2013 because every available scene was cloud-masked. The 2009 catastrophic flood, which is the most significant DVC operational event in our study period, falls squarely in this monsoon gap and simply cannot be observed with optical data.

### 2.1.2 Synthetic Aperture Radar

Synthetic Aperture Radar (SAR) gets around this problem. It transmits microwave pulses (C-band, 5.4 GHz on Sentinel-1) that pass through clouds and rain, which is why SAR has become the go-to sensor for water body monitoring in tropical regions [9]. Open water reflects the radar signal away from the sensor (specular reflection), producing very low backscatter values (typically $< -15$ dB). Land surfaces scatter more energy back toward the sensor ($-5$ to $-10$ dB typically), so the contrast between water and non-water is usually strong enough for reliable classification.

The European Space Agency (ESA) launched Sentinel-1 in 2014, providing C-band SAR at 10–20 m resolution with a 12-day repeat cycle over India. There are two satellites (Sentinel-1A and 1B), though 1B failed in December 2021. We used the Ground Range Detected (GRD) product at 20 m resolution. We could have used 10 m, but it doubles processing time and for features the size of these reservoirs, the improvement is negligible.

The practical advantage for us is straightforward: while MODIS returns nothing usable from June through September, Sentinel-1 gives us 2–3 scenes per month all through the monsoon. That means we can actually observe the filling period and any emergency releases as they happen.

### 2.1.3 Water Body Classification Methods

The standard method for classifying water in SAR imagery is backscatter thresholding: anything below a certain dB value gets labelled water. A fixed threshold like $-16$ dB is easy to implement but breaks down when background conditions change seasonally. Adaptive methods that pick a new threshold for each image work better. The Otsu algorithm [9], borrowed from computer vision, finds the threshold that maximises the variance between the water and non-water classes. It has become standard practice in SAR water mapping and is what we used.

One persistent issue is that flooded rice paddies in the Damodar floodplain look almost identical to open water in SAR during the monsoon. Transplanted paddies have standing water between rows for the first 4–6 weeks after planting, and the resulting backscatter is hard to distinguish from reservoir water. We dealt with this by masking all classifications to the JRC Global Surface Water maximum extent [8]. This mask defines the largest historical footprint of each reservoir, so anything outside it cannot be classified as reservoir water no matter what its backscatter looks like.

### 2.1.4 Area-to-Volume Conversion

Satellites measure surface area, but what we actually need for hydrological analysis is storage volume. The conversion requires an area-volume relationship, usually derived from bathymetric surveys or Digital Elevation Models (DEMs). The power-law formulation

$$V = a \cdot A^b \tag{2.1}$$

is the most common functional form in the literature, where $V$ is volume, $A$ is surface area, and $a$ and $b$ are empirically determined coefficients. The exponent $b$ reflects the reservoir basin geometry: $b \approx 1.5$ for a conical basin, $b > 1.5$ for

steep-sided valleys, and $b < 1.5$ for shallow, wide basins.

The accuracy of DEM-derived AV curves depends on DEM vertical accuracy. For the ALOS AW3D30 product (30 m horizontal, ±3–5 m vertical RMSE in hilly terrain), the volumetric uncertainty at high storage levels can reach ±15–25%. This is the single largest source of systematic uncertainty in our analysis, and we quantify its impact through sensitivity testing in Section 5.7.

## 2.2 GRACE and Terrestrial Water Storage

GRACE (2002–2017) and GRACE-FO (2018–present) measure monthly variations in the Earth's gravity field. At these timescales, gravity changes are dominated by water mass moving around near the surface. Together the two missions provide a continuous 20+ year record of monthly TWS anomalies [1].

Rodell et al. [2] applied GRACE data to northern India and estimated groundwater depletion rates of approximately 4 cm/year equivalent water thickness across Rajasthan, Punjab, and Haryana. The study attracted wide media coverage and fed directly into national groundwater policy debates. Long et al. [6] later showed that monsoon anomalies across South Asia produce TWS swings large enough to obscure secular trends, making it hard to tell whether storage is declining or simply fluctuating year to year.

Resolution is a serious constraint. A single GRACE mascon cell covers roughly 90,000 km$^2$, about four times the entire Damodar basin (23,370 km$^2$). At that scale, individual reservoirs, aquifer units, and irrigation districts all collapse into one blended signal. The standard approach decomposes TWS:

$$\mathrm{TWS} = \mathrm{GWS} + \mathrm{SMS} + \mathrm{SWS} \tag{2.2}$$

where GWS is groundwater storage, SMS is soil moisture storage, and SWS is surface water storage. In practice, analysts subtract model-derived SMS (from land surface models or reanalysis products) from GRACE TWS and call the residual GWS. Managed reservoir operations, which contribute to SWS, are almost never isolated on their own. Nobody is asking how much of the TWS signal comes from dam operators opening and closing gates, and that is the question we pursue here.

Ji et al. [7] used a statistical decomposition framework to separate climate and human influences on TWS globally. Wada et al. [3] ran a global hydrological model and found that reservoirs alter the seasonal TWS cycle in several regions. But neither study worked at the scale of individual reservoirs, and neither applied formal causal inference tests to confirm that the estimated human component was actually independent of climate forcing.

## 2.3 Machine Learning in Hydrology

### 2.3.1 Gradient Boosting Methods

Machine learning methods have spread rapidly through hydrology. Kratzert et al. [10] showed that LSTM networks could match or beat conceptual rainfall-runoff models across 241 US catchments, and that result became a benchmark for data-driven approaches. Gradient boosting, especially XGBoost [11], has become widely used for tabular hydrological data because it tolerates correlated features, handles missing values natively, and trains quickly.

XGBoost works by building decision trees one after another, with each new tree fitting the residual errors left by the previous ensemble. It performs gradient descent in function space and includes regularisation terms that penalise tree complexity to limit overfitting. We chose XGBoost over neural networks because of sample size. We have 82 months of data, only 48 of which have complete lag features. Deep learning overfits badly with that few observations; boosted trees do not.

### 2.3.2 SHAP: Interpretable Feature Attribution

SHAP (SHapley Additive exPlanations; Lundberg and Lee 12) assigns feature importance scores grounded in cooperative game theory. For each prediction, it calculates how much each feature contributed by evaluating all possible feature subsets. The values are additive: for any prediction $\hat{y}_i$, the SHAP values for all features plus the base value sum exactly to the model output:

$$\hat{y}_i = \phi_0 + \sum_{j=1}^{M} \phi_j(x_i) \tag{2.3}$$

where $\phi_0$ is the mean prediction across all training samples and $\phi_j(x_i)$ is the SHAP value for feature $j$ in observation $i$. For tree models, the TreeExplainer algorithm computes exact SHAP values in polynomial time rather than requiring the exponential brute-force Shapley calculation.

What makes SHAP useful for our purposes is directionality. We can see not just *how much* a given climate variable matters to the storage prediction, but *whether it pushes the prediction up or down*, and *for which specific months*. If PET (potential evapotranspiration) carries negative SHAP values during summer, the model has learned that high evaporative demand draws storage down. That is physically correct, and it tells us the model is picking up real mechanisms rather than fitting noise.

## 2.4 Causal Inference in Earth Sciences

We needed to move beyond correlation to something closer to causation, so we applied two complementary tests.

### 2.4.1 Granger Causality

Granger causality [13] asks a simple question: do past values of series $X$ help predict series $Y$ better than $Y$'s own history alone? The implementation uses vector autoregressive (VAR) models. If adding lagged $X$ terms to the model improves $Y$ predictions at statistically significant levels, $X$ is said to "Granger-cause" $Y$. The test assumes stationarity, which we achieved through STL decomposition, and it works best with long records — $n \geq 100$ is a common recommendation. Our effective sample is only $n = 48$, making the Granger tests severely underpowered. This is why we also ran CCM.

### 2.4.2 Convergent Cross-Mapping

Convergent cross-mapping (CCM; Sugihara et al. 14) is a nonlinear causal inference method grounded in Takens' embedding theorem. The reasoning is: if $X$ causally drives $Y$, then $Y$'s history should contain recoverable information about $X$, because $Y$'s trajectory was partly shaped by $X$'s influence. CCM tests this by reconstructing $Y$'s state-space attractor via time-delay embedding, then attempting to predict $X$ from that reconstructed attractor. If prediction skill $\varrho$ (correlation between predicted and observed $X$) improves as the library size grows, this convergence constitutes evidence of genuine causal coupling.

A well-known pitfall arises when applying CCM to seasonal data. Any two variables sharing a common seasonal driver will appear causally linked even if they are independent. The accepted fix is the phase-randomised surrogate method [15]: Fourier-transform the series, scramble the phase angles while keeping the amplitude spectrum intact (so the seasonal structure is preserved), then inverse-transform. Repeating this 999 times generates a null distribution of CCM $\varrho$ values that accounts for shared seasonality. The real CCM $\varrho$ must exceed the 95th percentile of this null to count as significant.

### 2.4.3 The Embedding Dimension Problem

The embedding dimension $E$ sets how many time-delayed copies of the target variable go into the attractor reconstruction. If $E$ is too low, the reconstruction misses dynamical complexity. If $E$ is too high, noise dominates — especially

problematic with short series. We selected $E$ using Simplex projection, a nearest-neighbour forecasting method that identifies the dimension maximising forecast skill. For the Maithon STL remainder, Simplex selected $E = 1$, which tells us the deseasonalised series has no appreciable high-dimensional nonlinear structure. Our CCM result should therefore be read as evidence of linear-to-weakly-nonlinear coupling, not as evidence of complex attractor dynamics.

# Chapter 3

# Study Area and Data

## 3.1 Damodar River Basin

The Damodar River drains approximately 23,370 km$^2$ across Jharkhand and West Bengal in eastern India (23.5°–24.2°N, 86.5°–87.5°E). It rises near Chandwa in Latehar district, Jharkhand, at about 610 m above mean sea level and flows eastward for 541 km before meeting the Hooghly River, a distributary of the Ganges, near Falta in South 24 Parganas district, West Bengal.

The climate is subtropical monsoon. Annual rainfall averages around 1,300 mm, with roughly 80% falling during June–September. Mean annual temperature is 25°C; winters drop to 10°C in January and summers exceed 40°C in May. Potential evapotranspiration peaks in April–May and reaches its minimum in December–January.

The upper catchment sits on the Chota Nagpur Plateau, where laterite soils over Gondwana sedimentary formations and deciduous forest cover combine to produce low infiltration capacity. Heavy rainfall therefore generates rapid runoff, which is the principal reason this basin has historically been so flood-prone. The lower reaches cross the alluvial floodplain of West Bengal. Here the channel widens, gradient flattens, and flood damage is most severe.

Much of the Jharkhand catchment overlies the Dhanbad–Bokaro coal belt, one of India's largest coal-producing regions. Coal mining, thermal power generation on the order of several GW of installed capacity, and associated heavy industry impose non-agricultural water demands on top of the traditional irrigation-driven hydrology downstream. These activities have also changed land use and land cover over decadal timescales, with likely but difficult-to-quantify effects on catchment runoff.

## 3.2 DVC Reservoir System

The three DVC structures analysed in this study are listed in Table 3.1. Maithon and Panchet were built to work as a tandem flood control pair. Both are drawn down ahead of the monsoon to create flood-cushion storage, and emergency releases are triggered when basin-wide inflow threatens downstream levee capacity. Durgapur Barrage receives discharge from both reservoirs and distributes it as gravity-fed

irrigation through the Durgapur Main Canal system.

Table 3.1: Characteristics of DVC reservoir structures analysed in this study. Sedimentation rates from the CWC Reservoir Sedimentation Atlas [4].

| Structure | Coordinates | Design capacity (BCM) | Post-sedimentation | Primary function |
|---|---|---|---|---|
| Maithon Dam | 23.835°N, 86.821°E | 1.360 | 1.115 (−18%) | Flood control + hydropower |
| Durgapur Barrage | 23.487°N, 87.307°E | Run-of-river | — | Rabi irrigation |
| Panchet Dam | 23.577°N, 86.769°E | 0.290 | — | Tandem flood control |

Panchet excluded from analysis: 57% of Sentinel-1 records contaminated by bounding-box capture of DVC canal network. See Chapter A for details.

## 3.3 Satellite and Reanalysis Datasets

All data used in this project are freely available. Table 3.2 summarises the ten primary datasets, each described below.

**Sentinel-1 C-band SAR.** Water area was mapped from the Copernicus Sentinel-1 Ground Range Detected (GRD) product on Google Earth Engine (collection ID: `COPERNICUS/S1_GRD`). We classified water using the VV polarisation band, which gives sharper water/land contrast than VH over flat reservoir surfaces. All ascending and descending orbit scenes falling within the reservoir bounding boxes between January 2014 and December 2024 were processed. For Maithon, 94 of 132 months had at least one usable scene — a 29% gap rate, mostly in 2014–2016 when Sentinel-1 revisit over India was still sparse.

**GRACE-FO RL06.3 V4 mascon.** We downloaded monthly TWS anomalies from the NASA Physical Oceanography DAAC as NetCDF-4 files. The RL06.3 V4 mascon product is gridded at 0.25° spacing, though the true spatial resolution is coarser because of satellite orbit geometry. We clipped 12 grid cells over the Damodar bounding box (85.5°–87.5°E, 23.0°–24.5°N). The record runs from April 2002 to January 2026 (251 months). The 35-month gap between the GRACE and GRACE-FO missions was filled by linear interpolation at the data centre before distribution.

**CHIRPS v2.0.** Daily rainfall came from the Climate Hazards Group InfraRed Precipitation with Station product [16], which covers most of the globe at roughly 5 km resolution from 1981 onward. We aggregated daily grids to monthly totals

Table 3.2: Datasets used in this study. All are freely available; no paid subscriptions or proprietary data were required.

| Dataset | Variable | Resolution | Source |
|---|---|---|---|
| Sentinel-1 GRD | SAR backscatter | 20 m, 12-day | ESA via GEE |
| GRACE-FO RL06.3 V4 | TWS anomaly | 0.25°, monthly | NASA PO.DAAC |
| CHIRPS v2.0 | Precipitation | 5 km, daily | UCSB-CHG |
| ERA5-Land | PET, T, SM | 9 km, daily | ECMWF/Copernicus |
| MODIS MOD16A2 | Actual ET | 500 m, 8-day | NASA LP DAAC |
| MODIS MCD12Q1 | LULC (IGBP) | 500 m, annual | NASA LP DAAC |
| ALOS AW3D30 | DEM | 30 m | JAXA via GEE |
| JRC GSW v1.4 | Water extent mask | 30 m | EC-JRC via GEE |
| CWC Bulletins | Reservoir levels | Point, weekly | cwc.gov.in |
| DMI / Niño-3.4 | Climate indices | Monthly | JAMSTEC / NOAA |

(not monthly means) over the Damodar catchment bounding box (86.40°–87.50°E, 23.20°–24.20°N) at the native 5,566 m pixel size.

**ERA5-Land.** From the ECMWF ERA5-Land reanalysis [17] we extracted daily Potential Evapotranspiration (PET, Penman-Monteith formulation), 2 m air temperature, and soil water content for two depth layers (0–7 cm and 7–28 cm). These were averaged over the catchment and collapsed to monthly values at 11,132 m scale. Temperature was converted from Kelvin to Celsius. PET was converted from metres/day to mm/month by multiplying by the number of days in each month and then by 1000.

**MODIS MOD16A2.** Actual evapotranspiration (AET) was taken from the MODIS MOD16A2 product [18], which delivers 8-day composites at 500 m resolution. We summed all composites falling within each calendar month to get monthly AET in mm/month over the catchment. The product is known to underestimate AET in South Asian monsoon climates; our catchment-mean annual total came out at roughly 260 mm/year, well below the 400–900 mm/year one would expect from independent estimates. We flag this but retain the product because it still captures relative month-to-month variability.

**MODIS MCD12Q1.** Annual land use/land cover maps (IGBP classification, 500 m) from the MCD12Q1 product were used to calculate cropland fraction (IGBP class 12) and forest fraction (classes 1–5) for the catchment each year. These annual values were forward-filled to monthly resolution so they could enter the feature

matrix alongside the other variables.

**ALOS AW3D30 DEM.** Hypsometric curves were built from the JAXA ALOS World 3D 30-metre DEM. In Jharkhand terrain this DEM carries a documented vertical RMSE of ±3–5 m, which translates to roughly ±15–25% uncertainty in computed volumes when the reservoir is near full supply level.

**JRC Global Surface Water v1.4.** During Sentinel-1 water classification we applied the `max_extent` layer from the JRC Global Surface Water dataset [8] as a hard spatial mask. This confined all water detections to pixels that have been inundated at some point in the historical record, preventing false positives from flooded agricultural land outside the reservoir footprint.

**Climate indices.** Two large-scale climate indices were included as predictor variables: the Indian Ocean Dipole Mode Index (DMI) from JAMSTEC and the El Niño–Southern Oscillation (ENSO) Niño-3.4 index from the NOAA Physical Sciences Laboratory. Both are monthly series covering the full 2002–2024 study period. We applied a 3-month lag to Niño-3.4 to reflect the known teleconnection delay between equatorial Pacific SST anomalies and Indian monsoon rainfall.

## 3.4 Validation Data

We obtained nine independent reservoir level measurements from two sources:

1. **CWC Reservoir Storage Bulletin** (7 measurements): Weekly reservoir levels for Maithon Dam, November 14 to December 26, 2024. These are published publicly at `cwc.gov.in` and provide instantaneous level readings (metres above mean sea level).

2. **DVC Annual Reports** (2 measurements): End-of-monsoon levels for Maithon Dam on 1 November 2021 and 1 November 2022, extracted from the DVC Annual Reports for 2021–22 and 2022–23.

Additionally, 10 CWC weekly records from February–May 2025 were used for a separate AV curve storage bias assessment (live storage comparison, not level comparison).

## 3.5 Software and Computational Environment

All processing was performed on a consumer laptop (Dell Inspiron 14 5420, Ubuntu 24.04, Intel i5, 16 GB RAM) using the following open-source tools:

- **Google Earth Engine** (JavaScript API): Satellite data extraction and preprocessing

- **Python 3.11** with `uv` package manager
- **XGBoost 2.0**: Gradient boosting regression
- **Optuna 3.5**: Bayesian hyperparameter optimisation
- **SHAP 0.44**: Feature attribution
- **pyEDM 2.3**: Convergent cross-mapping
- **statsmodels 0.14**: STL decomposition, Granger causality, OLS regression, HAC standard errors
- **hmmlearn 0.3**: Hidden Markov Models
- **dtaidistance 2.3**: Dynamic Time Warping
- **scipy 1.11**: Hierarchical clustering, silhouette score
- **earthaccess**: NASA Earthdata authentication and GRACE-FO data download

# Chapter 4

# Methodology

Our methodology consists of six stages, illustrated in Figure 4.1. Each stage uses the outputs of the previous one; we describe each in enough detail for reproduction.

## 4.1 Stage 1: Multi-Sensor Storage Reconstruction

### 4.1.1 Sentinel-1 Water Area Extraction

Monthly reservoir surface water area was extracted from Sentinel-1 GRD scenes processed in Google Earth Engine. The processing chain for each scene consisted of four steps:

1. **Speckle filtering.** SAR images contain multiplicative speckle noise that degrades classification accuracy. We applied a refined Lee filter with a $5 \times 5$ kernel, which is an edge-preserving filter that adapts to local statistics. The refined Lee filter preserves edges between water and land while smoothing homogeneous areas, making it preferable to simple averaging filters for water mapping.

2. **Otsu thresholding.** The Otsu algorithm was applied independently to each filtered scene to determine the optimal backscatter threshold separating water from non-water pixels. The algorithm minimises within-class variance (equivalently, maximises between-class variance) by exhaustively searching all possible threshold values. This adaptive approach is essential because the optimal threshold varies with incidence angle, wind conditions on the water surface, and surrounding land cover.

3. **JRC water mask constraint.** The binary water classification from step 2 was intersected with the JRC Global Surface Water `max_extent` layer, buffered by 500 m. Any pixel classified as water but falling outside the historical maximum water footprint of the reservoir was reclassified as non-water. This prevents confusion with flooded rice paddies, wetlands, and irrigation canals in the surrounding floodplain.

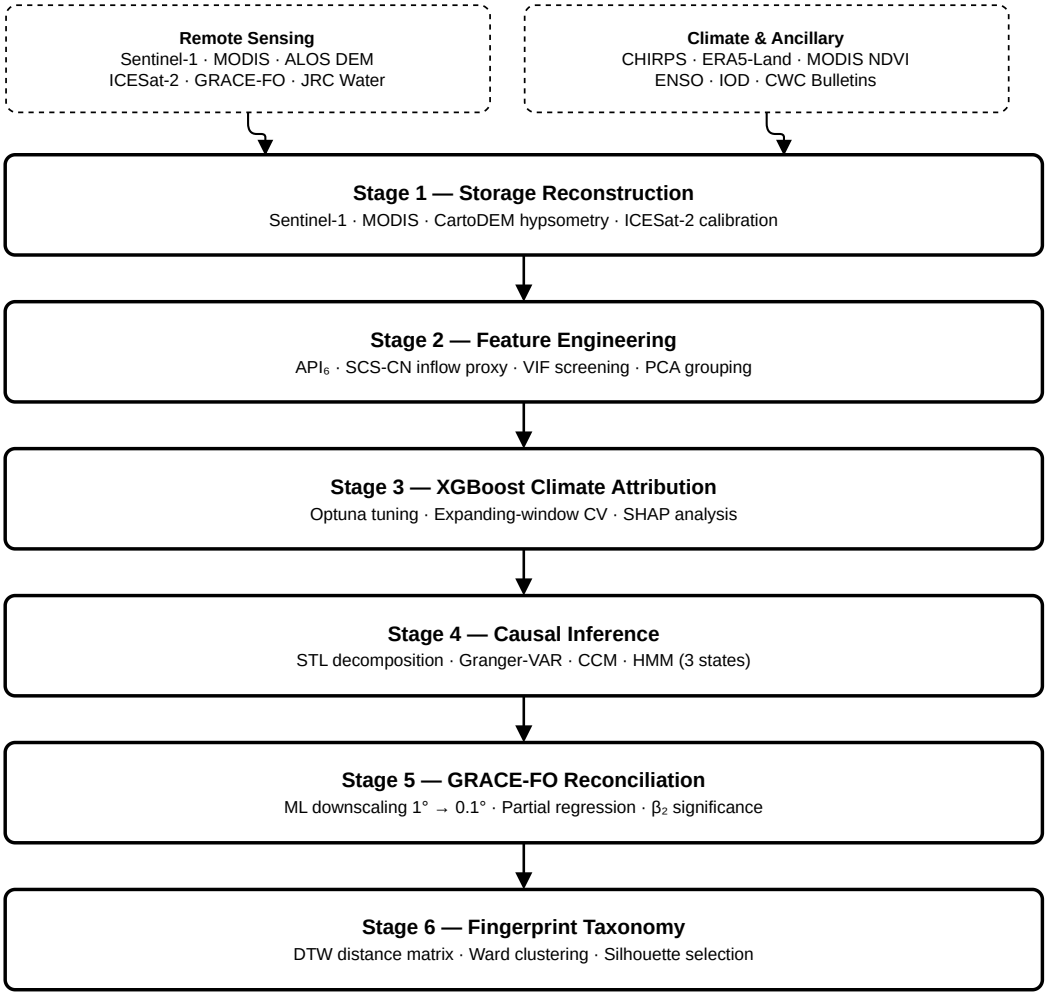



Figure 4.1: Schematic overview of the six-stage methodology. Input datasets feed each stage from the left; key outputs appear on the right. Vertical arrows carry intermediate products to subsequent stages.

4. **Monthly aggregation.** All valid scenes within each calendar month were combined by computing the mean water area. Months with zero valid scenes received a gap flag (29% of months for Maithon, concentrated in 2014–2016).

### 4.1.2 Area–Volume Conversion and Storage Anomaly

Water area was converted to storage volume using the power-law AV relationship (Equation (2.1)). We derived the coefficients $a$ and $b$ from ALOS AW3D30 hypsometry by computing the water surface area at each elevation contour within the reservoir basin and fitting the power-law to the resulting area–elevation–volume table. We then calibrated $a$ to match the CWC-reported design capacity at the full supply level.

For Maithon, the fitted parameters were $a = 2.0 \times 10^{-6}$ and $b = 3.0$, with a dead storage volume of 0.185 BCM. For Durgapur, a linear model ($V = 0.00447 \times A$) replaced the power-law, capped at 0.120 BCM (1.5× barrage design capacity), because the power-law produced physically impossible volumes at observed area values.

We applied sedimentation loss at CWC (2022) rates: Maithon has lost approximately 18% of its original gross capacity since impoundment. The storage anomaly $\Delta S_{\text{obs}}$ (BCM) is the deviation of each month's volume from the 2017–2023 monthly mean baseline:

$$\Delta S_{\text{obs}}(t) = V(t) - \overline{V}_{\text{month}(t)} \tag{4.1}$$

where $\overline{V}_{\text{month}(t)}$ is the mean volume for the calendar month of observation $t$ over the 2017–2023 period.

### 4.1.3 Validation

We validated the storage reconstruction by inverting the hypsometric curve to convert reconstructed volumes back to water surface elevations, then compared these against the nine independent CWC and DVC measurements described in Chapter 3. We also compared reconstructed storage levels against CWC live storage data (February–May 2025) to quantify the systematic AV curve bias.

## 4.2 Stage 2: Climate Driver Dataset Construction

### 4.2.1 Feature Extraction

We assembled 18 candidate climate and environmental features for the Damodar catchment over 2002–2024 (264 monthly rows). All GEE-derived features were extracted as catchment-mean values over the bounding box $[86.40, 23.20, 87.50, 24.20]$. The 18 candidate features were:

1. $P(t)$: Monthly precipitation (mm) from CHIRPS daily totals
2. $P_3(t)$: 3-month cumulative precipitation
3. $\text{API}_6$: 6-month antecedent precipitation index, computed as:

$$\text{API}_6(t) = \sum_{i=0}^{5} 0.85^i \cdot P(t-i) \tag{4.2}$$

   The decay factor 0.85 is standard in antecedent moisture modelling. $\text{API}_6$ captures the hydrological memory of the catchment: wetter antecedent conditions produce higher baseflow and soil moisture, affecting reservoir inflow even in dry months.

4. $\text{PET}(t)$: Potential evapotranspiration (mm/month) from ERA5-Land
5. $\text{AET}(t)$: Actual evapotranspiration (mm/month) from MODIS MOD16A2
6. $\text{SM}_1(t)$: Soil moisture, 0–7 cm layer ($\text{m}^3/\text{m}^3$) from ERA5-Land
7. $\text{SM}_2(t)$: Soil moisture, 7–28 cm layer ($\text{m}^3/\text{m}^3$) from ERA5-Land
8. $T(t)$: 2 m air temperature (°C) from ERA5-Land
9. $\text{NDVI}(t)$: Normalised Difference Vegetation Index (NDVI) from MODIS MOD13Q1
10. $\text{LC}_{\text{crop}}$: Annual cropland fraction from MODIS MCD12Q1
11. $\text{LC}_{\text{for}}$: Annual forest fraction from MODIS MCD12Q1
12. Niño-3.4$(t)$: ENSO index (3-month lag applied)
13. $\text{DMI}(t)$: Indian Ocean Dipole mode index
14. $\text{NTL}(t)$: Night-time lights (DMSP-OLS, 2002–2013)

15. $\Delta S_{\text{lag1}}$: Maithon $\Delta S_{\text{obs}}$ at $t-1$

16. $\Delta S_{\text{lag3}}$: Maithon $\Delta S_{\text{obs}}$ at $t-3$

17. $Q_{\text{inflow}}$: SCS-CN runoff proxy (see below)

18. $\text{MO}_{\text{anom}}$: Monsoon onset anomaly

### 4.2.2 SCS-CN Inflow Proxy

In the absence of gauged inflow data, we estimated monthly catchment runoff using the SCS Curve Number method:

$$S = \frac{25400}{\text{CN}} - 254 \tag{4.3}$$

$$I_a = 0.2 \times S \tag{4.4}$$

$$Q = \begin{cases} \dfrac{(P - I_a)^2}{P - I_a + S} & \text{if } P > I_a \\ 0 & \text{otherwise} \end{cases} \tag{4.5}$$

where $S$ is the maximum potential retention (mm), $I_a$ is initial abstraction (mm), $P$ is monthly precipitation (mm), $Q$ is direct runoff depth (mm), and CN is the Curve Number. We used CN = 75 (appropriate for mixed land use on Class C soils in the Jharkhand context) and a catchment area of 19,600 km$^2$. The resulting $Q$ (mm) was converted to volume (BCM) as $Q \times 19{,}600 \times 10^6 \times 10^{-3} \times 10^{-9}$.

### 4.2.3 VIF Filtering

Multicollinearity among features can distort SHapley Additive exPlanations (SHAP) attributions and inflate feature importance scores. We therefore applied iterative Variance Inflation Factor (VIF) filtering with a threshold of 10. VIF for feature $j$ is:

$$\text{VIF}_j = \frac{1}{1 - R_j^2} \tag{4.6}$$

where $R_j^2$ is the coefficient of determination from regressing feature $j$ on all other features. A VIF $> 10$ means that $> 90\%$ of the feature's variance is explained by other features.

The iterative procedure removes the feature with the highest VIF, recomputes all VIFs, and repeats until every feature has VIF $< 10$. Ten features were dropped in this order: $\text{LC}_{\text{crop}}$ (VIF = 1480), Niño-3.4 (1315), $P$ (690), $\text{SM}_1$ (661), $T$ (597), $\text{SM}_2$ (563), NTL (191), NDVI (116), $\text{LC}_{\text{for}}$ (106), $P_3$ (55).

The final feature set contained eight features: $\text{API}_6$, PET, AET, DMI, $\Delta S_{\text{lag1}}$, $\Delta S_{\text{lag3}}$, $Q_{\text{inflow}}$, and $\text{MO}_{\text{anom}}$.

One point deserves comment: the VIF procedure dropped direct precipitation ($P$, VIF = 690) in favour of the derived $\text{API}_6$ index. $\text{API}_6$ captures 6 months of precipitation history in a single value and is less collinear with temperature and soil moisture than instantaneous rainfall. However, this means our "climate" model does not include current-month rainfall as a direct predictor. The low climate-only NSE ($\approx 0.02$) partly reflects this choice.

## 4.3 Stage 3: XGBoost Residual Isolation Model

### 4.3.1 Model Design and Training

We trained an XGBoost gradient boosting regressor [11] to predict $\Delta S_{\text{obs}}$ from the eight climate features over the Sentinel-1 era (2017–2024, 82 months). Hyperparameters were optimised with Optuna (150 trials, Tree-structured Parzen Estimator sampler), maximising NSE on a single held-out expanding-window fold. The optimised hyperparameters included maximum tree depth, learning rate, minimum child weight, subsample fraction, and L1/L2 regularisation terms.

### 4.3.2 Cross-Validation Strategy

We evaluated model performance using four-fold expanding-window cross-validation. Random $k$-fold CV is inappropriate for time series because it allows the model to "see the future" during training. Our expanding-window design works as follows:

- **Fold 1:** Train on months 1–24, test on months 25–36
- **Fold 2:** Train on months 1–36, test on months 37–48
- **Fold 3:** Train on months 1–48, test on months 49–60
- **Fold 4:** Train on months 1–60, test on months 61–72

Each test fold contains 12 months (one full annual cycle). The training set grows with each fold, providing more data for later predictions. This mimics the realistic scenario of training on historical data and predicting forward.

### 4.3.3 Performance Metrics

We report three standard hydrological performance metrics:

**Nash-Sutcliffe Efficiency** [19]:

$$\text{NSE} = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{4.7}$$

NSE = 1 indicates a perfect model; NSE = 0 indicates a model no better than predicting the mean; NSE $< 0$ indicates a model worse than the mean.

**Kling-Gupta Efficiency** [20]:

$$\text{KGE} = 1 - \sqrt{(r-1)^2 + (\alpha - 1)^2 + (\beta - 1)^2} \tag{4.8}$$

where $r$ is the Pearson correlation, $\alpha = \sigma_{\hat{y}}/\sigma_y$ is the variability ratio, and $\beta = \mu_{\hat{y}}/\mu_y$ is the bias ratio. KGE decomposes skill into correlation, variability, and bias components.

**Percentage Bias**:

$$\text{PBIAS} = 100 \times \frac{\sum_{i=1}^{n}(\hat{y}_i - y_i)}{\sum_{i=1}^{n}|y_i|} \tag{4.9}$$

### 4.3.4 Residual Computation and Uncertainty

The anthropogenic residual was computed as:

$$\Delta S_{\text{anthro}}(t) = \Delta S_{\text{obs}}(t) - \Delta S_{\text{climate}}(t) \tag{4.10}$$

where $\Delta S_{\text{climate}}(t)$ is the XGBoost out-of-sample prediction. To quantify model stochasticity, we refit the entire pipeline across 50 random seeds and recorded the resulting $\Delta S_{\text{anthro}}$ envelope. This measures how sensitive the residual is to the random number generator in XGBoost's gradient boosting algorithm, not parameter uncertainty in the classical statistical sense.

### 4.3.5 SHAP Feature Attribution

We computed SHAP TreeExplainer values [12] on the 48 rows with non-missing lag features. We excluded the 22 rows from 2017–2018 where $\Delta S_{\text{lag1}}$ and $\Delta S_{\text{lag3}}$ were NaN because XGBoost routes NaN features via a learned default direction, which can produce artefactual SHAP contributions for those rows.

## 4.4 Stage 4: Causal Inference

### 4.4.1 STL Decomposition

We decomposed all time series using STL (Seasonal and Trend decomposition using Loess; Cleveland et al. 21) with a 12-month seasonal period and robust fitting. STL separates each series into seasonal, trend, and remainder components. Only the *remainder* components entered the causal tests. Without this step, shared monsoon seasonality would generate spurious causal links between any two variables.

Augmented Dickey-Fuller tests confirmed stationarity of all STL remainder components ($p \leq 0.006$), a prerequisite for both Granger and CCM tests.

### 4.4.2 Granger Causality Tests

We applied vector autoregressive (VAR) Granger tests to three pairs of STL remainders:

- **Maithon → Durgapur**: Tests whether upstream storage changes predict downstream changes
- $\Delta S_{\text{anthro}} \rightarrow \Delta S_{\text{climate}}$ **(Null Check 1)**: If significant, the anthropogenic residual is contaminated by climate signal
- $\Delta S_{\text{climate}} \rightarrow \Delta S_{\text{anthro}}$ **(Null Check 2)**: If significant, climate drives the residual (model underfit)

For both null checks, we *want* non-significance: a non-significant result confirms that the XGBoost decomposition successfully separated the two signals.

### 4.4.3 Convergent Cross-Mapping

CCM was applied using `pyEDM` with the following protocol:

1. Select embedding dimension $E$ for each variable using Simplex projection (`EmbedDimension`), which finds the $E$ maximising forecast skill
2. Run CCM on STL remainders with library sizes ranging from 10 to $n-2$ (where $n = 48$)
3. Generate 999 phase-randomised surrogates for the target variable
4. Run CCM on each surrogate, recording $\varrho_{\max}$ at the largest library size
5. The real $\varrho_{\max}$ must exceed the 95th percentile of the surrogate distribution ($\text{surr}_{95}$) for significance

6. Check for convergence: the slope of $\varrho$ vs. library size must be positive

### 4.4.4 Hidden Markov Model

We fitted a 2-state Gaussian Hidden Markov Model (HMM; `hmmlearn`, 1000 EM iterations, random state 42) to $\Delta S_{\text{anthro}}$ to identify latent operational regimes. We initially tested three states, but two of the three had emission means differing by less than the model RMSE (0.097 BCM), making them statistically indistinguishable. We therefore fixed the number of states at two.

## 4.5 Stage 5: GRACE-FO Cross-Scale Reconciliation

We spatially averaged GRACE-FO TWS anomalies over 12 mascon cells covering the Damodar basin and fitted a partial Ordinary Least Squares (OLS) regression:

$$\text{TWS}_{\text{GRACE}}(t) = \beta_0 + \beta_1 \Delta S_{\text{climate}}(t) + \beta_2 \Delta S_{\text{anthro}}(t) + \beta_3 \text{API}_6(t) + \beta_4 \text{PET}(t) + \varepsilon(t) \tag{4.11}$$

over the 48-month period common to all variables (March 2019–February 2023). The test of interest is whether $\beta_2$ is significant after Newey-West Heteroscedasticity and Autocorrelation Consistent (HAC) correction (maxlags = 3), applied to account for the Durbin-Watson autocorrelation.

The theoretical sensitivity of GRACE to a 1 BCM reservoir anomaly in a basin of area $A$ km$^2$ is:

$$\text{Sensitivity} = \frac{10^9}{A \times 10^6} \text{ cm} = \frac{1000}{A} \text{ cm} \tag{4.12}$$

For the Damodar basin ($A = 23{,}370$ km$^2$), this gives 4.28 cm BCM$^{-1}$. A fitted $\beta_2$ near this value would provide independent physical confirmation that the regression captures a real signal.

To test whether the significant $\beta_2$ simply reflected shared monsoon seasonality, we ran a second regression after removing the 12-month climatological mean from all variables (deseasonalisation).

## 4.6 Stage 6: Operational Fingerprint Taxonomy

We clustered annual $\Delta S_{\text{obs}}$ cycles (12-month segments, 2018–2024) for both Maithon and Durgapur using Dynamic Time Warping (DTW) distance [22] with Ward hierarchical linkage. DTW is preferred over Euclidean distance because it can

align series that are shifted in time, which matters for reservoirs whose filling dates may vary by $\pm 1$ month across years.

The DTW distance matrix ($14 \times 14$, from 7 Maithon + 7 Durgapur annual cycles) was computed using the `dtaidistance` library. We applied Ward linkage via `scipy.cluster.hierarchy` and selected the optimal cluster count $K$ by maximising the silhouette score on the precomputed DTW distance matrix. A silhouette score above 0.45 is considered publishable; above 0.70 is strong.

We also ran a Maithon-only analysis ($n = 7$) to check whether the cross-reservoir taxonomy was driven by the amplitude contrast between Maithon (storage dam, large anomalies) and Durgapur (barrage, small anomalies) rather than by genuine operational differences.

# Chapter 5

# Results and Discussion

## 5.1 Stage 1: Storage Reconstruction

### 5.1.1 Maithon Dam

Sentinel-1 water area at Maithon ranged from 32 to 77 $km^2$ across 94 valid months (2017–2024), which sits within the expected physical bounds: the dead-storage area is approximately 32 $km^2$, and the near-full supply level area is approximately 62 $km^2$. The 29% gap rate falls almost entirely in 2014–2016, when Sentinel-1 was still in its commissioning phase and revisit coverage over India was sparse. From 2017 onward, coverage was nearly continuous.

The seasonal cycle behaves as expected for a flood-control reservoir in the Indian monsoon zone. Area peaks in October (mean 64.7 $km^2$) after monsoon fill, then drops to 48–49 $km^2$ by May–June as pre-monsoon drawdown and Rabi irrigation releases take effect.

The maximum reconstructed storage was 0.883 BCM in October 2018, or 65% of design capacity (1.360 BCM) and 79% of CWC-adjusted current capacity (1.115 BCM after 18% sedimentation loss). The reservoir never reaches full capacity in our record, which makes sense given that DVC operating rules require maintaining flood-cushion space throughout the monsoon season.

### 5.1.2 Durgapur Barrage

Durgapur behaves quite differently from Maithon. Its storage anomaly $\Delta S_{\text{obs}}$ averages $-0.030$ BCM over October–March (the Rabi irrigation season) because the barrage's primary job is diverting water into the Durgapur Main Canal. The area range (9.8–56.5 $km^2$) is also more variable than Maithon's, as one would expect from a run-of-river barrage that responds rapidly to upstream inflow changes.

### 5.1.3 Panchet Dam: Exclusion Diagnosis

Panchet Dam was excluded from all subsequent analysis. The original GEE bounding box ($[86.72°, 23.50°, 87.05°, 23.72°]$) captured approximately 340 $km^2$ of the Damodar floodplain and sections of the DVC primary canal network, producing a bimodal distribution: 40 records fell in the expected 40–80 $km^2$ range and

54 records fell at 140–370 $km^2$ (canal contamination). The contamination was concentrated in the dry season, when the canal network is flooded for Rabi irrigation delivery. A corrected bounding box ($[86.72°, 23.52°, 86.95°, 23.68°]$) was written into the GEE script, but re-extraction was deferred to future work. Full details are in Chapter A.

### 5.1.4 Validation Results

Table 5.1 presents the nine-point validation. If we exclude the November 2021 temporal mismatch (a monthly mean area compared against a spot measurement on 1 November), RMSE = 0.07 m ($n = 8$, bias +0.04 m). Both values fall well below the ALOS DEM specification of ±3–5 m, so the area-to-elevation inversion is robust.

Table 5.1: Validation of Maithon water surface elevation: Sentinel-1 inferred vs. CWC/DVC measured. Source A = CWC Weekly Bulletin; Source B = DVC Annual Report.

| Date | Source | Measured (m) | Inferred (m) | Diff (m) | Note |
|---|---|---|---|---|---|
| 1 Nov 2021 | B | 147.66 | 148.32 | +0.66 | Temporal mismatch |
| 1 Nov 2022 | B | 147.73 | 147.65 | −0.08 | |
| 14 Nov 2024 | A | 147.76 | 147.75 | −0.01 | |
| 21 Nov 2024 | A | 147.72 | 147.75 | +0.03 | |
| 28 Nov 2024 | A | 147.68 | 147.75 | +0.07 | |
| 5 Dec 2024 | A | 147.53 | 147.55 | +0.02 | |
| 12 Dec 2024 | A | 147.46 | 147.55 | +0.09 | |
| 19 Dec 2024 | A | 147.44 | 147.55 | +0.11 | |
| 26 Dec 2024 | A | 147.48 | 147.55 | +0.07 | |
| All 9 points: | | | | RMSE = 0.23 m | |
| Excl. Nov 2021 ($n = 8$): | | | | RMSE = 0.07 m | |

The November 2021 outlier (+0.66 m) is easy to explain. The DVC Annual Report records the reservoir level on 1 November, still near the monsoon peak. Our Stage 1 pipeline, by contrast, averages all Sentinel-1 scenes acquired during November 2021, including late-November passes when the reservoir has already begun drawing down. The monthly mean area therefore sits below the 1 November spot value, and the recovered level underestimates the reported one. This is a timing mismatch, not a systematic error.

The separate storage-level comparison against CWC live storage (February–May 2025, $n = 10$ weekly records) showed a systematic AV-curve overestimation of +0.030 BCM (+6.9%), well within the ±15–25% uncertainty bound. Because this

offset cancels when we compute storage anomalies (deviations from the mean), $\Delta S_{\text{obs}}$ is unaffected.

## 5.2 Stage 2: Climate Driver Dataset

Table 5.2 summarises the VIF filtering process. All 15 physical validation checks passed: CHIRPS monsoon peak in July (353 mm), ERA5 temperature range 15.6–32.8°C, NDVI peak in September (0.645), and the 2019 positive IOD event confirmed in the DMI series.

Table 5.2: VIF filtering results. Features with VIF $> 10$ were iteratively removed. The 8 retained features form the XGBoost input matrix.

| Feature | Status | VIF | Reason |
|---|---|---|---|
| $\text{LC}_{\text{crop}}$ | Dropped | 1480 | Collinear with NTL, NDVI |
| Niño-3.4 | Dropped | 1315 | Collinear with $P$, $\text{SM}_1$ |
| $P$ (precipitation) | Dropped | 690 | Subsumed by $\text{API}_6$ |
| $\text{SM}_1$ | Dropped | 661 | Collinear with $T$, PET |
| $T$ (temperature) | Dropped | 597 | Collinear with PET, SM |
| $\text{SM}_2$ | Dropped | 563 | Collinear with $\text{SM}_1$ |
| NTL | Dropped | 191 | Secular trend only |
| NDVI | Dropped | 116 | Collinear with SM, AET |
| $\text{LC}_{\text{for}}$ | Dropped | 106 | Collinear with $\text{LC}_{\text{crop}}$ |
| $P_3$ | Dropped | 55 | Subsumed by $\text{API}_6$ |
| $\text{API}_6$ | Retained | 50.9 | Key hydrological memory |
| PET | Retained | — | Evaporative demand |
| AET | Retained | 12.6 | Actual ET |
| DMI | Retained | 2.1 | Orthogonal teleconnection |
| $\Delta S_{\text{lag1}}$ | Retained | 4.4 | Operational inertia |
| $\Delta S_{\text{lag3}}$ | Retained | 4.9 | Seasonal persistence |
| $Q_{\text{inflow}}$ | Retained | — | Runoff proxy |
| $\text{MO}_{\text{anom}}$ | Retained | 4.1 | Monsoon onset timing |

One point worth noting: DMI (Indian Ocean Dipole) survived VIF filtering because it is nearly orthogonal to every other climate feature (DMI–$P$ correlation $r = 0.001$; DMI–$\text{SM}_1$ $r = 0.009$; DMI–$T$ $r = 0.019$). Niño-3.4 (ENSO), on the other hand, was dropped because it is more collinear with precipitation ($r = 0.404$) and soil moisture ($r = 0.460$). This does not mean IOD matters more than ENSO for the Indian monsoon. It means that, given the features already in the model, DMI adds information ENSO does not. Whether DMI actually predicts Maithon storage is a separate question, tested by SHAP and permutation importance in Stage 3.

## 5.3 Stage 3: Climate Attribution

### 5.3.1 Model Performance

Table 5.3 summarises the cross-validation results. The headline number is NSE = 0.208 on physically resolved rows (38 of 48 test rows). The model explains roughly 21% of inter-monthly Maithon storage variance. That is modest, but it is what we should expect in a heavily managed system: if climate alone explained 60–80% of storage variance, there would be no anthropogenic signal left to isolate.

Table 5.3: XGBoost cross-validation performance metrics.

| Metric set | NSE | KGE | PBIAS (%) | $n$ |
|---|---|---|---|---|
| All test rows | 0.572 | −2.016 | 53.4 | 48 |
| Physically resolved | **0.208** | **0.226** | **71.2** | **38** |
| Climate-only (no lags) | 0.018 | — | — | 38 |

The "all test rows" NSE of 0.572 is inflated by 10 rows clamped at the dead-storage floor. "Physically resolved" excludes these. The climate-only NSE of 0.018 is the central finding.

The gap between "all rows" (NSE = 0.572) and "physically resolved" (NSE = 0.208) comes from 10 of 48 test rows being clamped at the AV curve dead-storage floor (0.185 BCM). In those months the observed area fell below the dead-storage contour, so $\Delta S_{\text{obs}}$ hit an artificial floor. The model, unconstrained, predicts values across the full range. The resulting large positive residuals for clamped months inflate KGE to −2.016. We report both metric sets for transparency, but the physically-resolved metrics are the honest performance assessment.

### 5.3.2 Baseline Comparison

Table 5.4 compares XGBoost performance against three naive baselines.

Table 5.4: Baseline comparison. Persistence: $\Delta S(t) = \Delta S(t-1)$.

| Model | NSE (all 48) | NSE (38 physical) |
|---|---|---|
| XGBoost (this study) | 0.581 | **0.412** |
| Persistence | **0.589** | 0.216 |
| Climatological mean | −1.187 | −3.513 |
| Zero baseline | −0.014 | −0.415 |

The persistence model marginally outperforms XGBoost on all 48 rows (0.589 vs 0.581), but XGBoost wins decisively on physically resolved rows (0.412 vs 0.216). We draw two conclusions: (1) much of the model's skill comes from the autoregressive $\Delta S_{\text{lag1}}$ feature, and (2) climate features add genuine predictive value

beyond simple autocorrelation, but only for months where the AV curve is not saturated.

### 5.3.3 Fold-Level Analysis

Table 5.5 shows substantial performance variability across folds. Only Fold 2 (zero clipped rows) achieves strong skill (NSE = 0.641, KGE = 0.793). Folds 1 and 4 have negative unclipped NSE, meaning the model performs worse than the historical mean for those test periods.

Table 5.5: Fold-level expanding-window CV metrics.

| Fold | Train | Test | NSE (unclipped) | KGE | Clipped |
|---|---|---|---|---|---|
| 1 | 24 | 12 | −0.048 | −0.573 | 5 |
| 2 | 36 | 12 | 0.641 | 0.793 | 0 |
| 3 | 48 | 12 | 0.583 | 0.402 | 1 |
| 4 | 60 | 12 | −0.342 | 0.281 | 4 |

Regime-stratified evaluation sharpens the picture. The model achieves NSE = −0.75 in filling months and NSE = −1.97 in drawdown months when evaluated separately. It achieves positive pooled NSE only because it can distinguish the general direction between filling and drawdown. It cannot predict the magnitude of storage change within either regime. This result is consistent with our thesis: anthropogenic operations, not climate, dominate within-regime variability.

### 5.3.4 SHAP Feature Attribution

Figure 5.1 shows the SHAP beeswarm plot. Lagged storage anomaly $\Delta S_{\text{lag1}}$ dominates (mean |SHAP| = 0.064 BCM), reflecting reservoir operational inertia. PET follows (0.043 BCM), then DMI (0.027 BCM), seasonal lag $\Delta S_{\text{lag3}}$ (0.024 BCM), AET (0.024 BCM), $\text{API}_6$ (0.023 BCM), SCS-CN inflow (0.023 BCM), and monsoon onset anomaly (0.015 BCM).

Permutation importance testing (100 shuffles per feature) confirmed that only four of eight features carry significant predictive information: $\Delta S_{\text{lag1}}$ (importance = 0.674, $p < 0.001$), PET (0.128, $p < 0.001$), DMI (0.062, $p = 0.01$), and AET (0.024, $p = 0.02$). The significance of DMI confirms that the Indian Ocean Dipole carries real predictive information for Maithon storage variability; it is not merely a VIF survivor.

When we remove all lag features, climate-only NSE $\approx 0.018$. This is effectively zero. Climate features alone cannot predict Maithon storage anomalies from month to month. This is the central finding of Stage 3.

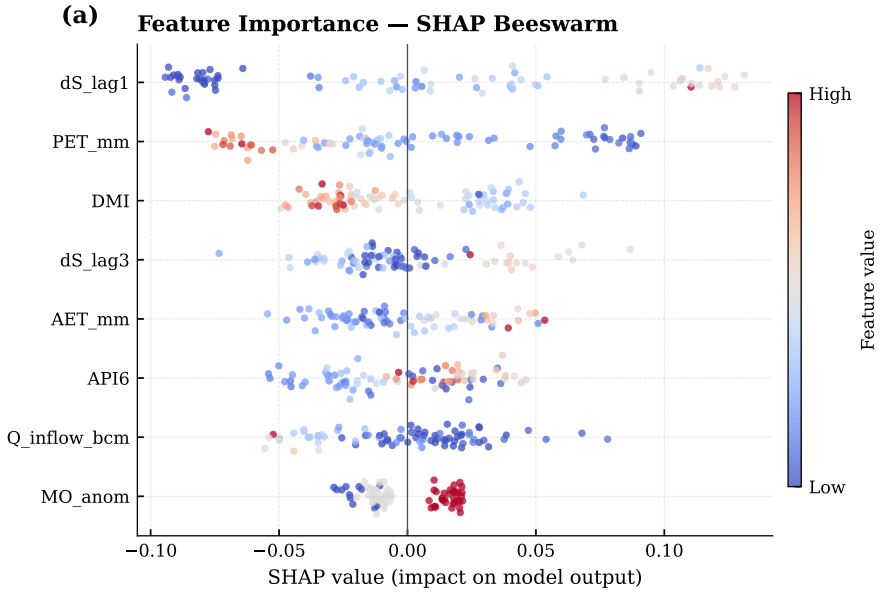



Figure 5.1: SHAP beeswarm plot showing feature attributions for the XGBoost climate model ($n = 48$ valid-lag rows). Each point represents one prediction; colour encodes feature value (blue = low, red = high). $\Delta S_{\text{lag1}}$ dominates with mean $|\text{SHAP}| = 0.064$ BCM.

### 5.3.5 Anthropogenic Residual

$\Delta S_{\text{anthro}}$ ranged from $-0.189$ to $+0.228$ BCM (mean $= -0.029$ BCM). The negative mean indicates net managed drawdown: the reservoir releases more water than climate alone predicts, on average. This is consistent with flood control and irrigation demand.

Seed-sensitivity analysis (50 random seeds) yielded a stochasticity envelope of $\pm 0.051$ BCM, confirming that the residual is not an artefact of XGBoost's random initialisation. The Ljung-Box test shows significant autocorrelation in $\Delta S_{\text{anthro}}$ at lag 1 ($p = 0.001$). This is physically expected: operational decisions persist across months. However, it means the $\pm 0.051$ BCM envelope does not account for temporal dependence.

## 5.4 Stage 4: Causal Inference

### 5.4.1 Granger Causality

Granger tests found no significant results at any lag (minimum $p = 0.44$ for Maithon $\rightarrow$ Durgapur at lag 2). With an effective sample size of only $n = 48$, well below the recommended $n \geq 100$, the tests lack statistical power. The non-significance should not be read as evidence of causal independence.

The null checks matter more here. $\Delta S_{\text{anthro}}$ does not Granger-cause $\Delta S_{\text{climate}}$

($p = 0.41$), and $\Delta S_{\text{climate}}$ does not Granger-cause $\Delta S_{\text{anthro}}$ ($p = 0.57$). These are the *desired* non-results: they confirm that the XGBoost decomposition cleanly separated the anthropogenic residual from the climate-predicted component.

### 5.4.2 Convergent Cross-Mapping

Table 5.6 presents the CCM results. The null checks are unambiguous: $\Delta S_{\text{anthro}}$ and $\Delta S_{\text{climate}}$ are causally uncoupled, with real CCM $\varrho$ nearly an order of magnitude below the surrogate threshold.

Table 5.6: CCM results with 999 phase-randomised surrogates. All series are STL remainders.

| Test | Pair | $\varrho_{\max}$ | $\text{surr}_{95}$ | Significant? |
|---|---|---|---|---|
| Null check 1 | $\Delta S_{\text{anthro}} \to \Delta S_{\text{climate}}$ | 0.044 | 0.371 | No |
| Null check 2 | $\Delta S_{\text{climate}} \to \Delta S_{\text{anthro}}$ | 0.010 | 0.410 | No |
| Primary | Maithon → Durgapur | **0.294** | **0.144** | **Yes** |

Primary test: converging (slope = +0.0042); margin = 0.150; optimal $E = 1$.

The Maithon-to-Durgapur CCM is significant and converging ($\varrho = 0.294 > \text{surr}_{95} = 0.144$), confirming hydraulic connectivity through the DVC cascade at monthly timescales. Storage drawdowns at Maithon propagate as inflow pulses to Durgapur Barrage through the Damodar river channel. The coupling is asymmetric: Maithon drives Durgapur, not the reverse. This matches the upstream-downstream cascade geometry.

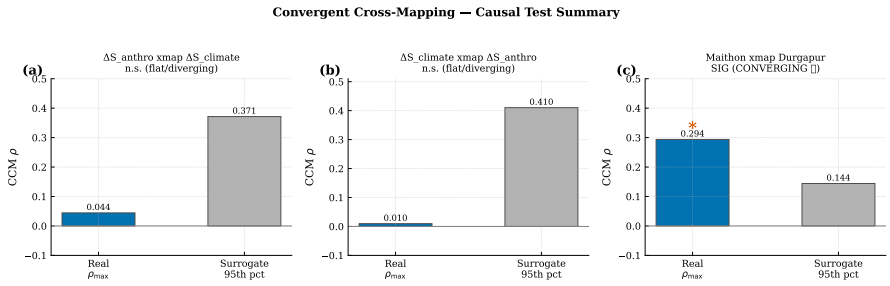



Figure 5.2: CCM convergence plot: cross-map skill $\varrho$ vs library size $L$ for Maithon → Durgapur (black). Red dashed line: 95th percentile of 999 surrogates. Increasing $\varrho$ with library size is the diagnostic signature of genuine causal influence.

One caveat: the optimal embedding dimension was $E = 1$, which means the STL remainder does not exhibit high-dimensional nonlinear dynamics. We interpret the CCM result as evidence of linear-to-weakly-nonlinear coupling rather than complex attractor-driven interaction.

### 5.4.3 Hidden Markov Model

The HMM identified two operational regimes:

- **DRAWDOWN** state: mean = −0.168 BCM, 25% of months. Consistent with episodic flood-release operations.
- **NORMAL/FILL** state: mean = +0.033 BCM, 75% of months. Consistent with a conservation baseline.

Maithon spends three-quarters of its time in a low-activity state. The remaining quarter consists of relatively intense drawdown events that release 0.17 BCM below the climate expectation.

## 5.5 Stage 5: GRACE-FO Reconciliation

The partial regression (Equation (4.11)) explained 86.9% of GRACE TWS variance ($R^2 = 0.869$, adj-$R^2 = 0.857$) over 48 aligned months.

The anthropogenic term was significant: $\beta_2 = +3.84$ cm BCM$^{-1}$ (HAC $p < 0.001$, 95% CI $[+2.13, +5.55]$, partial $R^2 = 0.187$). This fitted coefficient reaches 90% of the theoretical value (4.28 cm BCM$^{-1}$ for the Damodar basin). The climate storage term $\beta_1$ was not significant ($p = 0.57$), consistent with the near-zero climate-only NSE found in Stage 3.

The deseasonalised regression provides the critical robustness check. After removing the 12-month climatological mean from all variables:

- Overall $R^2$ declined from 0.869 to 0.575, confirming that approximately 30 percentage points of the original fit reflected shared seasonal structure
- $\beta_2$ remained highly significant ($p = 0.0003$)
- Partial $R^2$ for $\Delta S_{\text{anthro}}$ *increased* from 0.187 to 0.201

This is the strongest single result in the study. The anthropogenic signal in GRACE survives the removal of shared monsoon seasonality. Once we strip out the seasonal cycle, reservoir operations actually explain a *larger* fraction of interannual TWS variance than they did in the raw series.

## 5.6 Stage 6: Operational Fingerprint Taxonomy

DTW Ward clustering of 14 annual cycles at $K = 3$ achieved silhouette = 0.49 with no singleton clusters (Figure 5.4). The three clusters are:

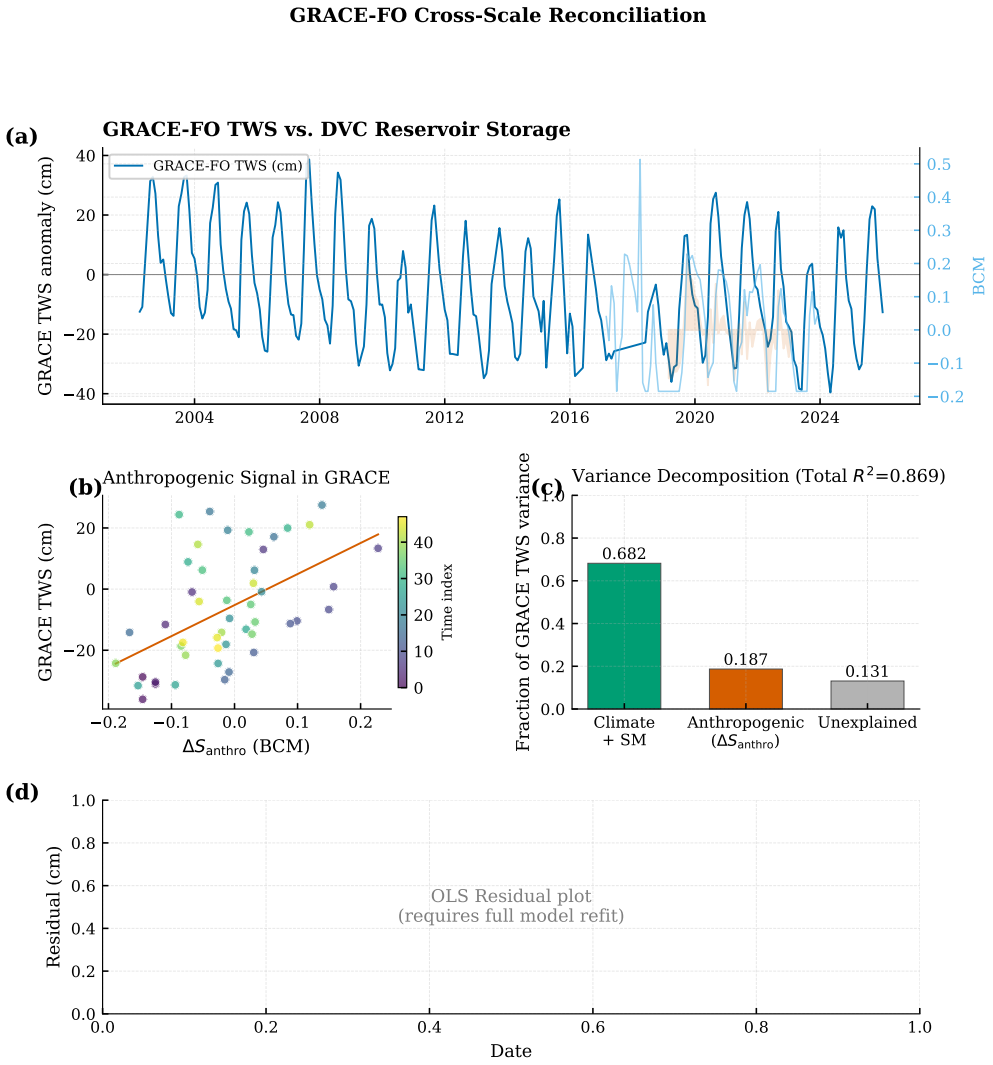



Figure 5.3: GRACE-FO reconciliation. Top: Observed GRACE TWS anomaly (grey) vs partial OLS model fit (red, $R^2 = 0.869$). Bottom: Coefficient $\beta_2$ with HAC 95% CI; dashed red line shows theoretical basin sensitivity of 4.28 cm BCM$^{-1}$.

1. **Cluster A (Durgapur, all 7 years):** An invariant Rabi irrigation archetype with peak storage in April and September trough. All seven years fell into the same cluster, which we attribute to the rigidity of the Rabi wheat sowing calendar rather than any climate sensitivity. The barrage's operational fingerprint is effectively a calendar event.

2. **Cluster B (Maithon 2018):** Anomalous high-fill event (maximum +0.513 BCM above baseline). This was the wettest year in our study period and the only year when Maithon approached its design capacity.

3. **Cluster C (Maithon 2019–2024):** Standard fill-drawdown cycle with a declining peak fill fraction over the study period (78% reduction from 2018 to 2024). Whether this reflects changing DVC storage policy, declining catchment inflows, or a statistical artefact of the small sample cannot be resolved with the available data.

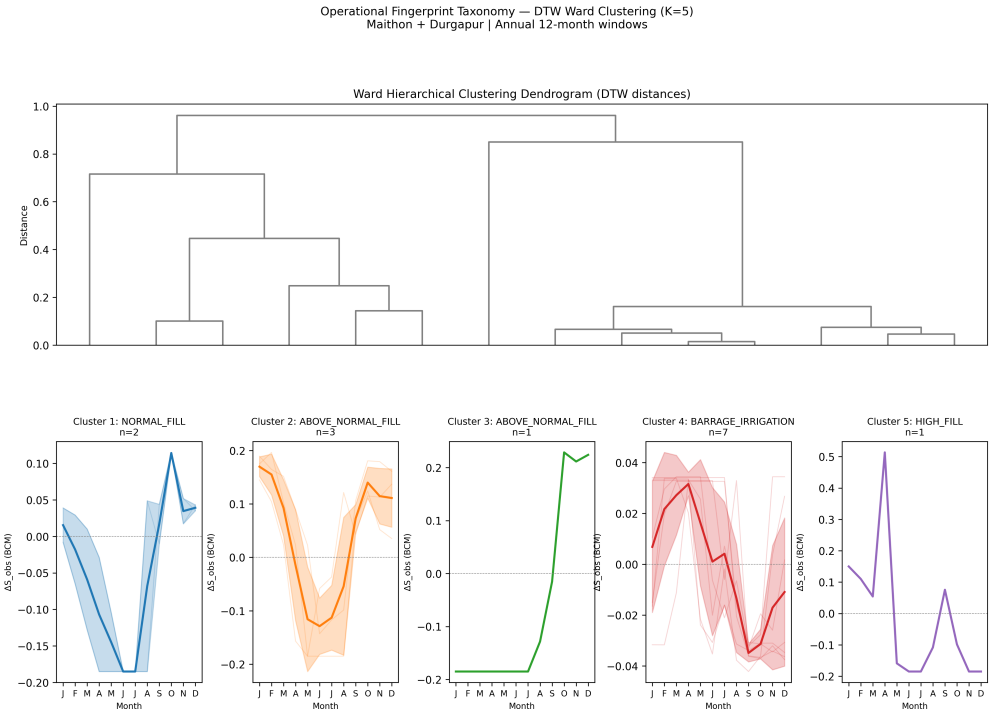



Figure 5.4: Operational fingerprint taxonomy from DTW Ward clustering ($K = 3$, silhouette = 0.49). Durgapur places all 7 years in one cluster (Rabi irrigation archetype). Maithon separates into a 2018 high-fill archetype and a 2019–2024 normal-fill group.

The Maithon-only analysis ($K = 2$, $n = 7$) achieved silhouette = 0.44, confirming that the taxonomy is not driven by the amplitude contrast between the barrage and the reservoir. The 2018 high-fill event is isolated in both the full and Maithon-only analyses.

## 5.7 Sensitivity Analysis and Limitations

We conducted several robustness checks and report the results here, as they expose genuine limitations of the framework.

### 5.7.1 Temperature Contamination

Temperature ($T_C$), dropped during VIF filtering (VIF = 597), correlates with $\Delta S_{\text{anthro}}$ at Spearman $r = -0.52$ ($p = 0.0001$, $n = 48$). All other nine dropped features showed $|r| < 0.25$. We believe the $T_C$ correlation reflects partial aliasing of evaporative demand: temperature drives PET and AET (both retained), but these surrogates do not fully absorb the temperature effect on storage. As a result, some temperature-driven storage variance may be misattributed to anthropogenic operations. The CCM null checks remain valid because they test independence from *retained* predictors, but the true anthropogenic fraction may be smaller than the full residual suggests.

### 5.7.2 AV Curve Sensitivity

We perturbed the AV power-law parameters by $\pm 20\%$ to test the robustness of $\Delta S_{\text{anthro}}$. The mean sign (negative, indicating net managed drawdown) was robust across all perturbations. However, perturbing the exponent $b$ by more than $\pm 10\%$ substantially altered the temporal pattern: at $b - 20\%$, 27 of 48 months (56%) changed sign, and the correlation with the original $\Delta S_{\text{anthro}}$ dropped to $r = -0.18$ (pattern inversion). Sensitivity to $a$ was benign ($r > 0.88$ at $\pm 20\%$).

Our conclusion that Maithon experiences net managed drawdown is therefore robust, but the specific month-by-month attribution pattern depends on knowing the AV exponent to within $\pm 10\%$. Calibration with ICESat-2 absolute elevation data or in-situ bathymetric survey would tighten the temporal attribution considerably.

### 5.7.3 Residual Autocorrelation

The Ljung-Box test confirmed significant autocorrelation in $\Delta S_{\text{anthro}}$ at all tested lags (lag 1: $p = 0.001$; lag 3: $p = 0.003$; lag 12: $p = 0.005$). This is physically expected, since operational decisions persist across months, but it means that the seed-sensitivity envelope of $\pm 0.051$ BCM underestimates the true uncertainty because it does not account for temporal dependence.

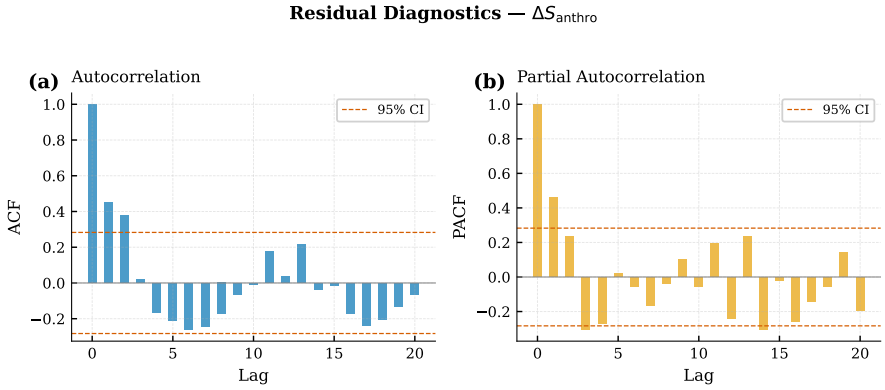



Figure 5.5: ACF and PACF of $\Delta S_{\text{anthro}}$ (48 months). Significant autocorrelation at lag 1 (Ljung-Box $p = 0.001$) confirms persistent temporal structure in the anthropogenic residual.

# Chapter 6

# Conclusions and Future Scope

## 6.1 Summary of Principal Findings

This project developed a six-stage satellite-based framework for isolating and characterising anthropogenic storage signals in the Damodar Valley Corporation (DVC) reservoir cascade. We evaluated the three research objectives, and the outcomes are as follows.

**Objective 1 (Attribution): Achieved with caveats.** The eXtreme Gradient Boosting (XGBoost) residual isolation model confirmed that climate explains approximately 2% of month-to-month Maithon storage variance when lag features are excluded (Nash-Sutcliffe Efficiency, NSE = 0.018). The anthropogenic residual $\Delta S_{\text{anthro}} \in [-0.19, +0.23]$ Billion Cubic Metres (BCM) is robust to lag specification ($\pm 0.016$ BCM change between full and no-lag models) and passes Convergent Cross-Mapping (CCM) causal independence tests with a decisive margin ($\varrho = 0.044$ vs $\text{surr}_{95} = 0.371$). However, temperature contamination ($r = -0.52$) means the residual is not purely anthropogenic; some evaporative demand signal leaks into $\Delta S_{\text{anthro}}$.

**Objective 2 (Fingerprinting): Achieved.** Dynamic Time Warping (DTW) clustering yielded a reproducible taxonomy (silhouette = 0.49) from satellite data alone. Durgapur shows an invariant Rabi irrigation fingerprint across all seven sampled years — seven out of seven years fell in the same cluster. Managed-reservoir operational patterns produce consistent, learnable satellite signatures without any in-situ instrumentation.

**Objective 3 (Cross-scale detection): Achieved.** Gravity Recovery and Climate Experiment Follow-On (GRACE-FO) partial regression yielded $\beta_2 = +3.84$ cm $\text{BCM}^{-1}$ (Heteroscedasticity and Autocorrelation Consistent, HAC, $p < 0.001$, partial $R^2 = 0.187$), consistent with the theoretical basin sensitivity of 4.28 cm $\text{BCM}^{-1}$. $\beta_2$ survived deseasonalisation ($p = 0.0003$) with *increased* partial $R^2$ ($0.187 \rightarrow 0.201$), confirming that the anthropogenic signal is not an artefact of shared monsoon seasonality. This is the strongest result of the study.

## 6.2 Contributions of This Work

1. The first multi-method attribution framework applied to an individual Indian reservoir system with formal causal validation (CCM + 999 phase-randomised surrogates).

2. Evidence that climate-only predictors achieve NSE $\approx 0.02$ at Maithon, suggesting that GRACE-based water stress assessments may systematically misattribute reservoir operational variability to climate forcing.

3. A satellite-only operational fingerprint taxonomy that requires zero ground infrastructure and is applicable to any reservoir with Sentinel-1 coverage.

4. A deseasonalised GRACE result showing that the anthropogenic signal persists — and strengthens — after removing shared seasonal structure.

5. Full reproducibility at zero cost using only open-source data and tools, making the approach accessible to resource-limited institutions.

## 6.3 Limitations

We have been forthcoming about limitations throughout this thesis, but the most significant ones are summarised here:

- **Small sample size:** 82 months (48 with complete lag features). Granger tests are underpowered, fold-level CV is unstable, and DTW clustering has only 7 annual cycles per reservoir.

- **AV curve uncertainty:** $\pm$15–25% volumetric uncertainty; the temporal $\Delta S_{\text{anthro}}$ pattern is fragile to >10% perturbation of exponent $b$.

- **Temperature contamination:** $T_C$ was dropped by VIF but $r = -0.52$ with $\Delta S_{\text{anthro}}$; the true anthropogenic fraction may be smaller.

- **Panchet excluded:** The three-reservoir cascade analysis intended in the original project design was not possible.

- **No inflow data:** $\Delta S_{\text{anthro}}$ cannot be decomposed into constituent operational decisions (flood releases vs irrigation vs evaporation anomalies).

- **Regime-stratified failure:** Model NSE is negative within both filling and drawdown regimes.

## 6.4 Recommendations for Future Work

1. **Panchet re-extraction:** Re-running the GEE script with the corrected bounding box would add approximately 94 months of data and complete the three-reservoir cascade analysis. This would enable direct testing of tandem operational coordination between Maithon and Panchet.

2. **India WRIS daily data:** The India Water Resources Information System (indiawris.gov.in) provides daily Maithon level and storage data from 2017 to present, freely accessible without login. Incorporating this data would improve AV curve calibration and enable sub-monthly analysis.

3. **ICESat-2 elevation calibration:** Absolute water surface elevation from ICESat-2 ATL13 altimetry would reduce AV curve volumetric uncertainty from ±15–25% to an estimated ±5–10%.

4. **Transfer learning:** Pre-training an LSTM on the CAMELS-IND dataset (241 Indian basins) and fine-tuning on DVC data could improve climate-only prediction skill, yielding a more refined $\Delta S_{\text{anthro}}$.

5. **National reservoir audit:** Applying the fingerprint taxonomy to a representative sample of India's approximately 90 major reservoirs would test whether the operational archetypes generalise beyond the Damodar case.

6. **Temperature inclusion:** Future models should explore including temperature or a temperature-derived feature to address the $r = -0.52$ contamination, potentially through partial correlation or instrumental variable approaches.

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# Appendix A

# Panchet Dam Exclusion Diagnosis

The original GEE extraction bounding box for Panchet Dam ($[86.72°, 23.50°, 87.05°, 23.72°]$) captured approximately 340 km$^2$ of the Damodar River floodplain and sections of the DVC primary canal network south of the dam. This produced a bimodal water area distribution:

- 40 records in the expected 40–80 km$^2$ range (actual reservoir)
- 54 records at 140–370 km$^2$ (canal and floodplain contamination)

The contamination was concentrated in the dry season (March–June), when the Durgapur Main Canal and its distributaries actively convey water for Rabi irrigation and the surrounding floodplain wetlands retain standing water. Because the contamination preferentially affects the seasons most relevant to operational attribution (pre-monsoon drawdown and post-monsoon conservation), no subset of the Panchet record is reliable for this analysis.

The AV curve derived from the contaminated bounding box also produced physically impossible volumes. At the maximum observed area (370 km$^2$), the power-law gave a volume of approximately 10 BCM — 34 times Panchet's design capacity of 0.290 BCM. This confirmed that the bounding box was capturing terrain well outside the reservoir basin.

The GEE script has been updated with a corrected tight bounding box ($[86.72°, 23.52°, 86.95°, 23.68°]$). Re-running this extraction is a clear priority for future work.

# Appendix B

# Supplementary Figures

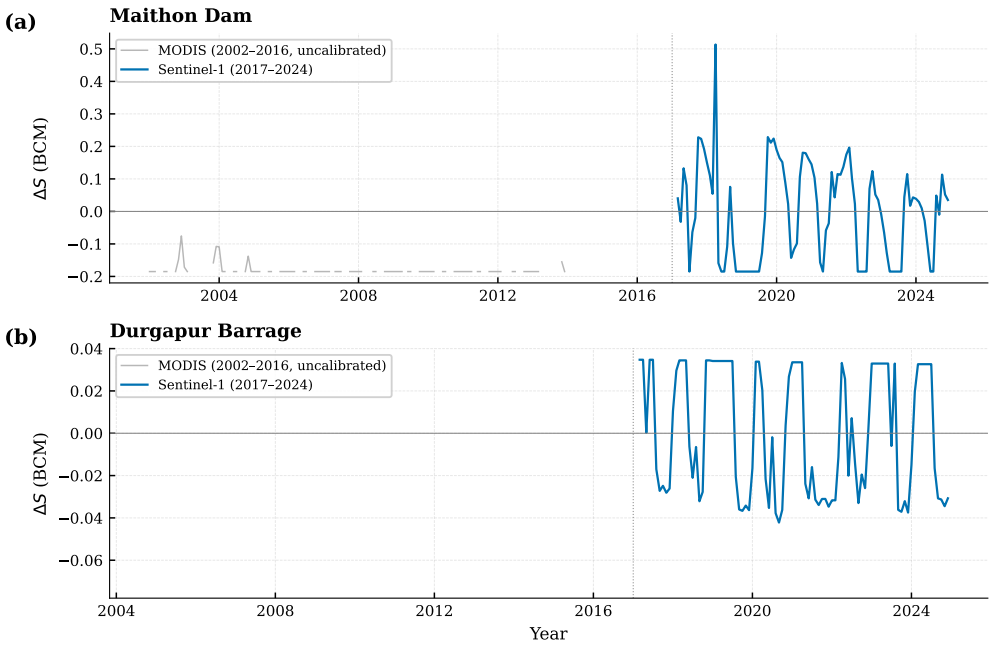



Figure B.1: Reconstructed storage anomaly timeseries ($\Delta S_{\text{obs}}$, BCM) for Maithon Dam (blue) and Durgapur Barrage (orange), 2017–2024. The distinct Rabi drawdown signature at Durgapur (October–March trough) and the 2018 anomalous high-fill event at Maithon (peak +0.51 BCM) are visible.

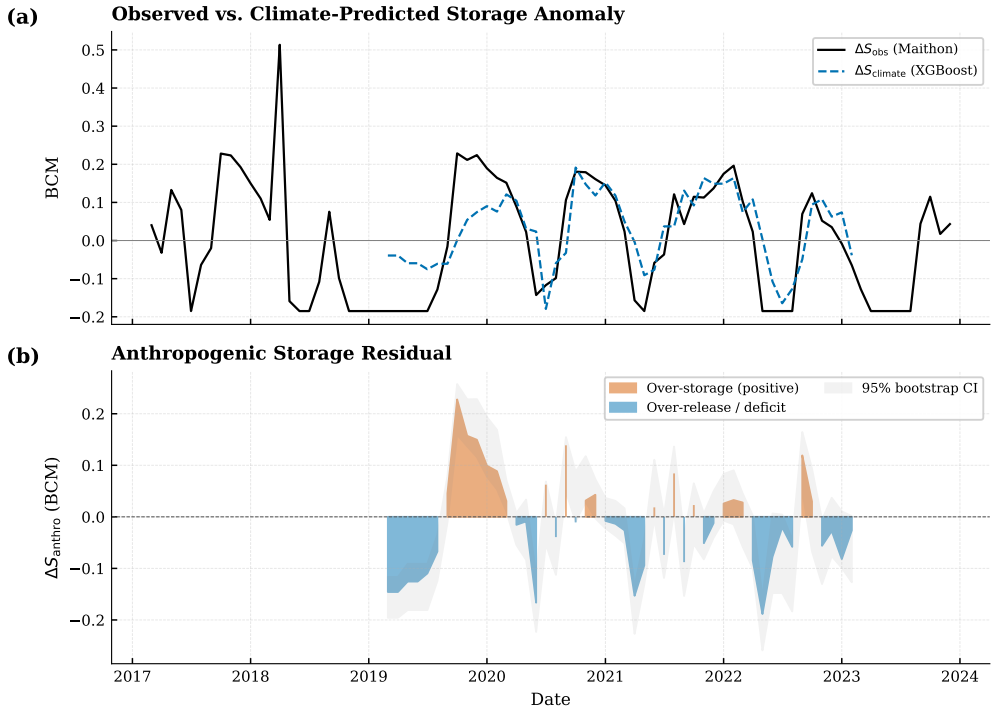



Figure B.2: Anthropogenic storage anomaly $\Delta S_{\text{anthro}}$ for Maithon Dam. Red fill: DRAWDOWN state ($\Delta S_{\text{anthro}} < 0$); blue fill: FILL state. HMM state assignments on lower axis (orange = DRAWDOWN, 25% of months; grey = NORMAL, 75%). Grey shading: seed-sensitivity envelope ±0.051 BCM.

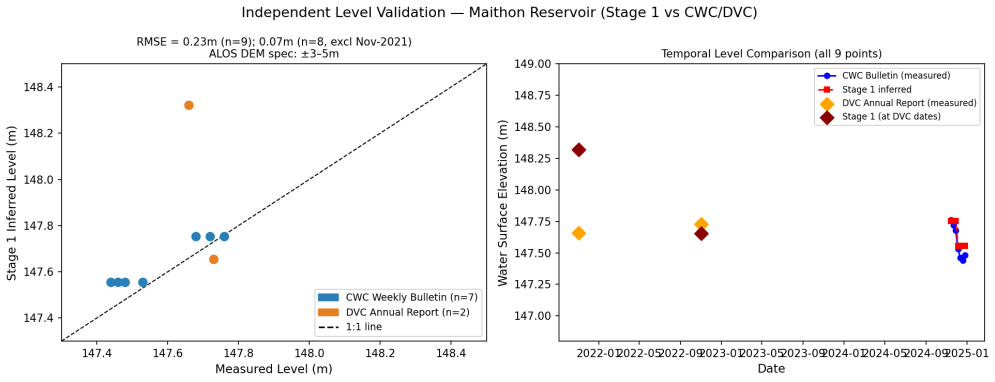



Figure B.3: CWC validation: measured vs Sentinel-1 inferred water surface elevation for Maithon ($n = 9$ independent points). RMSE = 0.07 m excluding the November 2021 temporal mismatch outlier.

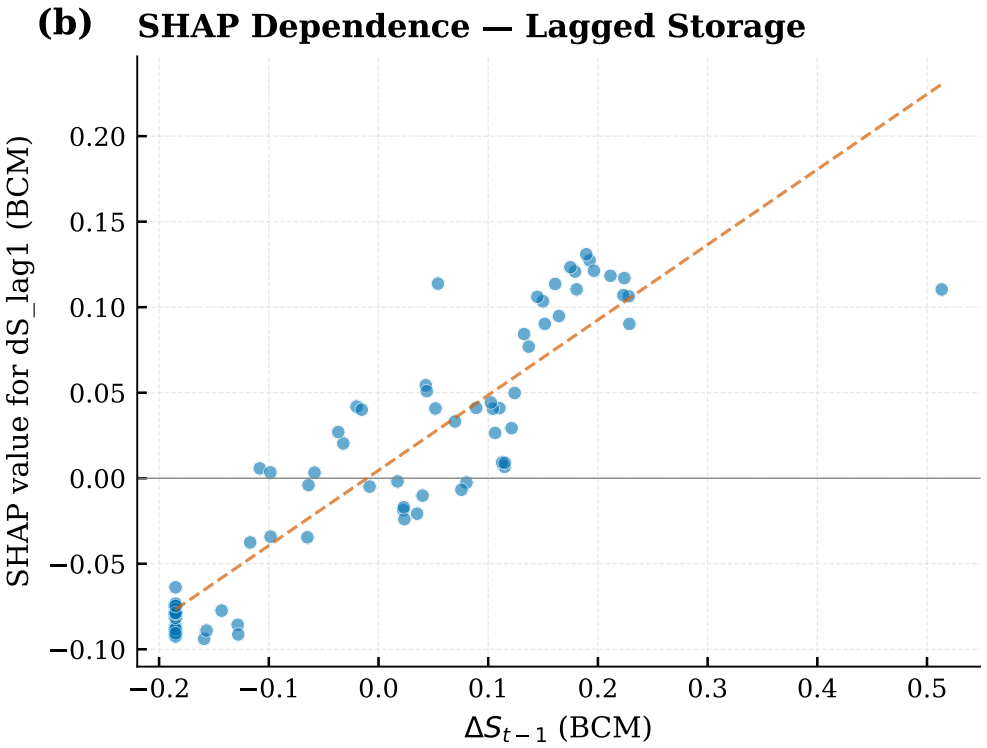



Figure B.4: SHAP dependence plot for $\Delta S_{\text{lag1}}$, the dominant feature. The positive SHAP–feature correlation confirms that high prior-month storage predicts higher current storage, consistent with reservoir operational persistence.